Indian Institute of Technology Jodhpur

M.Tech.
Program Course Curriculum
Department of Environmental and Sustainability Engineering (DESE)

1. Introduction

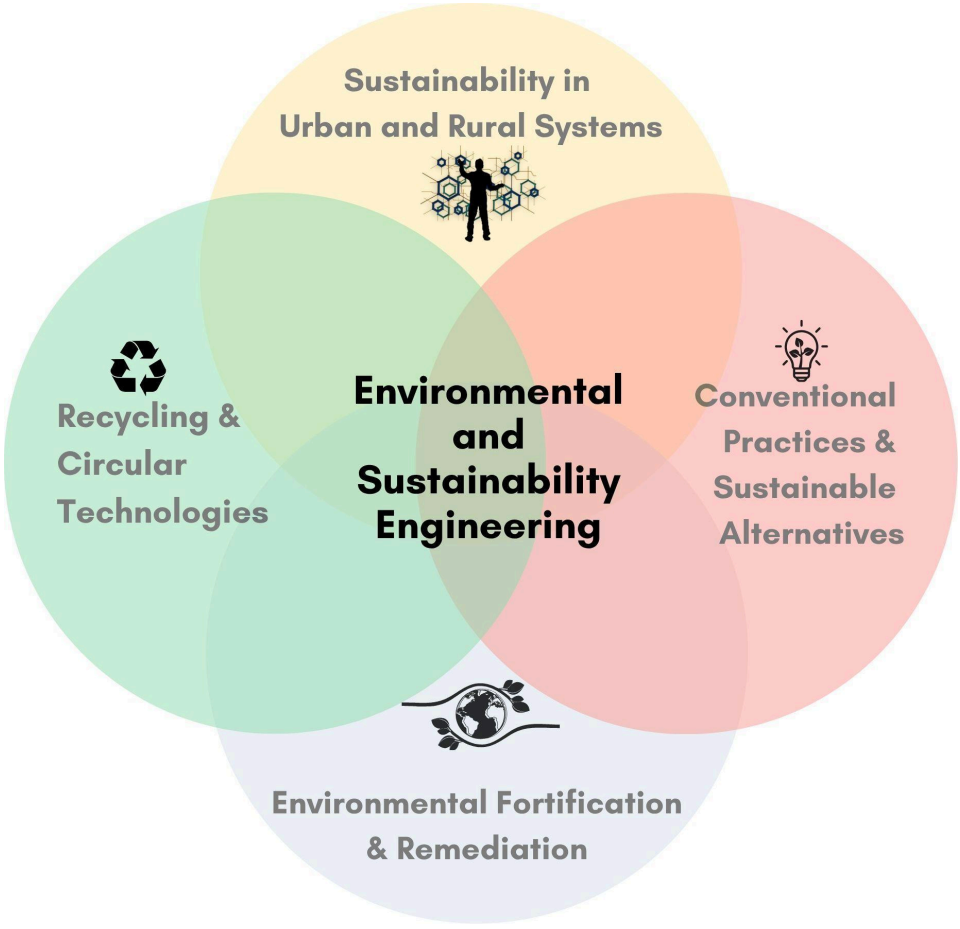
Sustainability has emerged as a defining national and global imperative, reshaping industrial transitions, development pathways, and public policy worldwide. Escalating challenges related to climate change, resource depletion, air pollution, environmental degradation, and socio-economic inequities increasingly influence decisions across sectors such as energy, water resources, food systems, waste management, and urban infrastructure. Addressing these interconnected challenges requires approaches that extend beyond disciplinary silos, integrating engineering innovation with environmental understanding, systems thinking, and socio-economic considerations.

India's commitment to achieving net-zero emissions by 2070 and reducing the emission intensity of its GDP by 45% by 2030 underscores a decisive shift toward low-carbon and resource-efficient development. Long-term assessments estimate that India's green economy could generate up to ~35 million green jobs by 2047, highlighting sustained employment potential in sustainability and climate-focused sectors. Realizing these ambitions requires professionals equipped with interdisciplinary sustainability expertise that integrates engineering, environmental science, data-driven decision-making, and policy awareness, capabilities that remain limited within conventional disciplinary training.

In this context, IIT Jodhpur occupies a unique and strategic position. Situated in the Thar Desert, one of India's most climate-vulnerable regions, the institute functions as a natural living laboratory for sustainability, where solutions can be developed, tested, and scaled under conditions of water scarcity, extreme heat, and fragile ecosystems. Building on this foundation, the M.Tech. in Environmental and Sustainability Engineering is envisioned as a flagship professional program that integrates engineering rigor with sustainability principles, indigenous knowledge systems, and emerging technologies. The program aims to develop Environmental and Sustainability Engineering with strong systems thinking, technical expertise in circular economy, energy transition, rural–urban sustainability, and environmental remediation, and the ability to design scalable, context-sensitive solutions for climate-resilient and low-carbon development, while contributing meaningfully to regional development, national priorities, and the United Nations Sustainable Development Goals (SDGs).

2. Objectives

The objective of MTech. in Environmental and Sustainability Engineering program is to produce industry-ready professionals with strong systems thinking and interdisciplinary engineering skills to address complex sustainability challenges across circular economy, energy transition, rural–urban systems, and environmental remediation. The program equips graduates with in-depth technical knowledge and hands-on experience in recycling and circular technologies, resource efficiency, waste engineering, decarbonization pathways, and sustainable alternatives to conventional energy practices. It enables students to design and implement context-sensitive solutions for water, air, soil, agriculture, infrastructure, and ecosystem restoration, integrating appropriate technologies, indigenous knowledge systems, and data-driven decision-making. Through this approach, the program prepares future engineers and leaders to contribute effectively to green industries, sustainable enterprises, policy implementation, and national and global sustainability goals, including the United Nations Sustainable Development Goals (SDGs).



3. Graduate Attributes

Graduates of the MTech. in Environmental and Sustainability Engineering will be able to:

1. Apply systems-oriented and transdisciplinary thinking to address complex sustainability challenges across circular economy, energy transition, rural–urban systems, and environmental remediation.
2. Design, implement, and manage sustainable technologies, infrastructure, and engineering systems suitable for diverse environmental, climatic, and socio-economic contexts.
3. Integrate engineering solutions with social, cultural, ecological, and indigenous knowledge considerations while engaging responsibly with communities and stakeholders.
4. Understand industry requirements, sustainability entrepreneurship, and professional practices relevant to green technologies and sustainable development.
5. Demonstrate ethical responsibility, teamwork, leadership, and inclusive collaboration in multidisciplinary and multi-stakeholder environments.
6. Contribute to regional, national, and global sustainability initiatives in alignment with Viksit Bharat 2047, the National Action Plan on Climate Change (NAPCC), and the United Nations Sustainable Development Goals (SDGs), respectively.

4. Learning Outcomes

Upon successful completion of the program, students will be able to:

1. Apply advanced Environmental and Sustainability Engineering principles to real-world challenges related to water, energy, waste, agriculture, infrastructure, and ecosystem management.
2. Use systems thinking, modeling approaches, AI, and data analytics for sustainability assessment, optimization, and decision support.
3. Design, evaluate, and optimize technologies, policies, and practices for circular economy, decarbonization, resource efficiency, and climate resilience.
4. Execute industry-oriented projects and a master’s thesis addressing practical sustainability problems in rural and urban contexts.
5. Communicate technical analyses and solutions effectively to academic, industry, government, and community stakeholders.

6. Translate technical knowledge into actionable outcomes through innovation, entrepreneurship, technology deployment, and policy engagement.

5. Job Opportunities & Workforce Demand

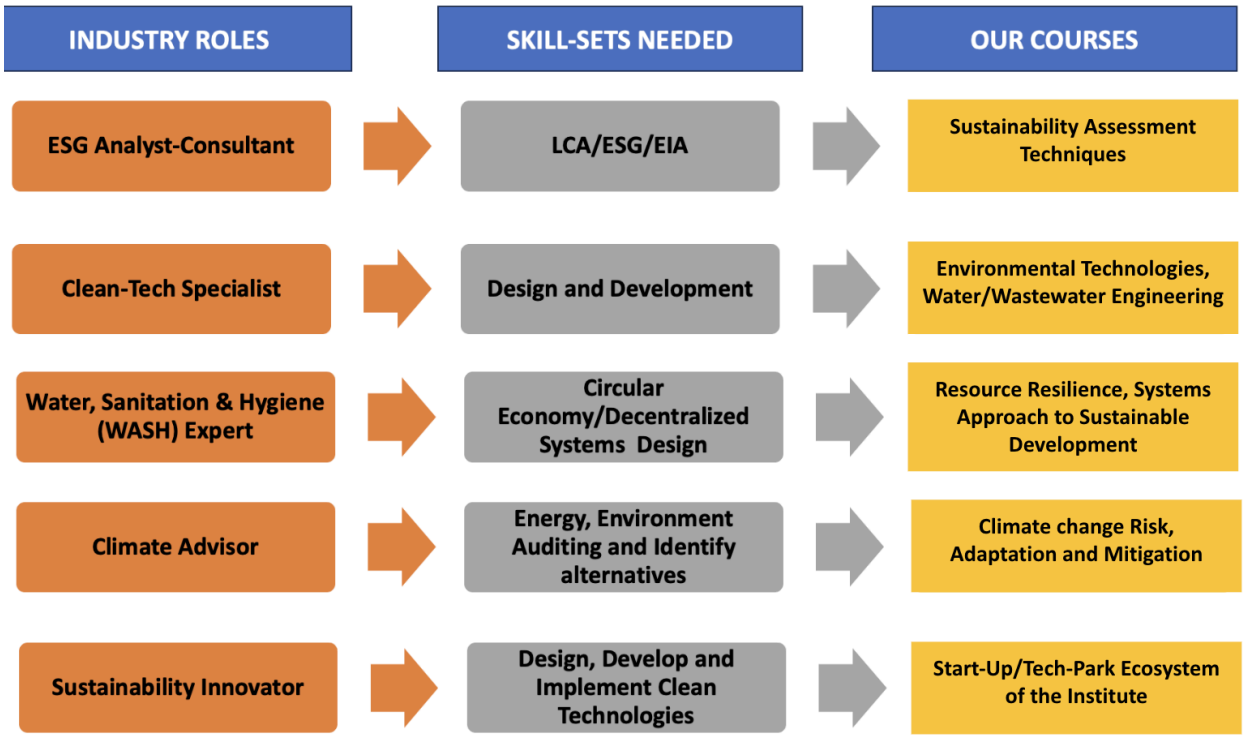
Global demand for sustainability and green-skills professionals is growing rapidly, with reports indicating ~12% year-on-year growth in green talent demand (2022–23) and an estimated ~24 million net new jobs worldwide linked to low-carbon and climate-resilient transitions across energy, infrastructure, industry, and environmental services. In India, only ~5.9% of the workforce is currently classified as green, highlighting a significant skill gap. Projections indicate the creation of ~7.3 million green jobs by 2027–28 and up to ~35 million jobs by 2047, driven by renewable energy, sustainable mobility, urban infrastructure, and environmental services. India's renewable energy workforce alone crossed ~1 million jobs in 2023, reflecting sustained sectoral expansion aligned with national net-zero commitments.

Graduates of the MTech. programme are expected to find opportunities across:

1. Industry & Corporate Sustainability / ESG / EHS
2. Sustainability & Climate Consulting
3. Energy Transition & Climate-Tech Sectors
4. Government, PSUs & Urban Infrastructure Agencies
5. International Organisations & NGOs
6. Research, Innovation & Doctoral Pathways

***Indicative Career Roles*:** Sustainability Analyst, ESG Consultant, Environmental Consultant, Decarbonization Analyst, Energy Systems Analyst, Policy Analyst, Sustainability Researcher, Urban/Community Sustainability Facilitator

6. Skill Sets and Career Pathway Alignment



7. Eligibility Criteria

	Qualifying Degree	**Requirement**
With Fellowship	A Bachelor's degree in Engineering (B.E.) or Bachelor of Technology (B.Tech.) in Civil Engineering/Chemical Engineering/ Environmental Science and Engineering/Agriculture Engineering/Mechanical Engineering/Materials and/or Metallurgical Engineering/Engineering Sciences/Biotechnology and/or Bio Engineering/Architecture and Planning or any other relevant engineering discipline.	Candidates with a valid GATE score in Civil, Environmental, Agriculture, Mechanical, Chemical, Materials, Biotechnology, Architecture, and Engineering Sciences.
	A Bachelor's degree in Engineering (B.E.) or Bachelor of Technology (B.Tech.) in Civil Engineering/Chemical Engineering/ Environmental Science and Engineering/Agriculture Engineering/Mechanical Engineering/Materials and/or Metallurgical Engineering/Engineering Sciences/Biotechnology and/or Bio Engineering/Architecture and Planning or any other relevant engineering discipline from any of the CFTIs with CPI or CGPA 8/10 or above.	The requirement of valid GATE score is exempted.

Without Fellowship	A Bachelor's degree in Engineering (B.E.) or Bachelor of Technology (B.Tech.) in Civil Engineering/Chemical Engineering/ Environmental Science and Engineering/Agriculture Engineering/Mechanical Engineering/Materials and/or Metallurgical Engineering/Engineering Sciences/Biotechnology and/or Bio Engineering/Architecture and Planning or any other relevant engineering discipline. **With minimum CPI or CGPA of 6/10**	Written test or interview

8. Student Intake

MoE Seats (Full Time Regular with Fellowship)	**Non MoE Seats** (Full Time Regular without Fellowship, Self Sponsored / Full Time Sponsored / Part Time / Part Time (Project Sponsored))
08	07

9. Semester-wise distribution of credits for M.Tech. students for the batches from Academic Year 2025-26.

I Semester						**II Semester**				
Cat	*Course Code*	*Course Title*	*L-T-P*	*Credits*		*Cat*	*Course Code*	*Course Title*	*L-T-P*	*Credits*
C	ETL7130	Sustainability Assessment Techniques	3-0-0	3		C	ETL7110	Systems Approach to Sustainable Development	3-0-0	3
C	ETL7140	Resource Resilience	3-0-0	3		C	ETL7120	Environmental Technologies	3-0-0	3
E	ETL7XXX	Elective	3-0-0	3		C	ETL7XXX	Environmental and Sustainability Engineering Laboratory	0-0-4	2
E	ETL7XXX	Elective	3-0-0	3		E	ETL7XXX	Elective	3-0-0	3
NG1	LAN7010	Technical Communication	1-0-0	S/U		E	ETL7XXX	Elective	3-0-0	3
						NG2	LAN7020	Ethics and Professional	1-0-0	S/U

								Life		
Total				**12**		*Total*				**14**
Summer Term										
T	ETP7XXX	Thesis		8						
Toal				**8**						
III Semester						**IV Semester**				
T	ETP7XXX	Thesis		16		T	ETP7XXX	Thesis		16
Toal				16			Total			16

10. Program Structure

S.No.	*Category*	*Course Category Title*	*Total Courses*	*Total Credits*
1	C	*COMPULSORY*	5	14
2	E	*ELECTIVES*	4	12
3	NG	*Non-Graded (2 credits)*	2	0
4	T	Thesis	3	40
Total				**66**

11. List of Elective Courses

1. Water Sustainability
2. Principles of Membrane Fabrication
3. Industrial Waste Management and Circular Economy
4. Introduction to ESG Regulation, Reporting, and Accounting
5. Technologies for Rural Sanitation
6. Biodiversity and Natural Resources
7. Climate change Risk, Adaptation and Mitigation
8. Industrial Ecology and Life Cycle Assessment

12. Detailed Course Contents

Program Compulsory Courses

Title	**Sustainability Assessment Techniques**	Number	ETL7XX1
Department	DESE	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech., M.Sc., M.S., M.Tech., Ph.D.	Type	Compulsory
Prerequisite	Basic math and data analysis knowledge is recommended. Pre-readings are provided to ensure smooth progression.	Course Level	700

Objectives

The instructor will:

1. Introduce key concepts in life cycle thinking, GHG accounting, and decarbonization strategies within a systems sustainability framework.
2. Demonstrate practical methods and tools for assessing environmental performance using LCA software, global databases, and ESG analytics platforms.
3. Guide students in applying data-driven approaches for climate decision-making, policy evaluation, and corporate sustainability reporting.

Learning outcomes

The students will be able to:

1. Conduct GHG accounting and life cycle assessments across sectors using relevant standards and digital tools.
2. Interpret sustainability metrics (e.g., GWP, ESG indicators) and simulate decarbonization scenarios using OpenLCA, R, and global datasets.
3. Apply decision-support methods (MCDA, sensitivity analysis) to evaluate sustainability trade-offs and model climate risks.

Contents

Materials Flow Analysis (MFA): Analysis, evaluation and design of anthropogenic systems (companies, cities, countries, the world); Carbon metrics and budgeting techniques; Scope 1, 2 & 3 accounting with case studies (institutional, industrial, organizational); Climate disclosure and reporting frameworks (CDP, TCFD, BRSR); Transboundary emissions and carbon leakage; Policy mechanisms and negotiation (UNFCCC, Paris Agreement); Applied decarbonization strategies [7 lectures]

Life Cycle Thinking: Concepts of cradle-to-grave and cradle-to-cradle; Goal and scope definition; Life Cycle Inventory (LCI); Life Cycle Impact Assessment (LCIA); Attributional vs. consequential LCA; Hybrid approaches; Sectoral case studies: EVs, cement, agri-food, ICT; Limitations and emerging issues [8 lectures]

Sustainability Tools & Analytics: Modelling of different environmental effects of products and services (climate change, toxicity, land use); Hands-on with LCA tools (OpenLCA, GaBi, GREET, SimaPro, Brightway2); Global databases (ecoinvent, USLCI, India GHG Platform, Agri-footprint); Hands-on assessments across sectors (green buildings, renewables, packaging, mobility); Integration with decision-support frameworks (SDG alignment, single-score vs. multi-criteria). [12 lectures]

Data to Decision: Sustainability indicators and frameworks; Data preprocessing and integration (Hands-on with R-based tools); Exploratory and predictive analytics; Climate risk modeling; Decision-making under uncertainty (MCDA, cost-benefit, scenario/sensitivity analysis); Capstone Activity: From Data to Decision [12 lectures]

Textbooks

1. Blokdyk, Gerardus (2019), GHG Emissions Accounting A Complete Guide 2019 Edition, Practical Tools for Self Assessment, ISBN-10 : 0655542779, ISBN-13 : 978-0655542773
2. Muthu, S. S. (Ed.). (2022). Greenhouse gas accounting: Global frameworks and emerging issues. Springer. https://doi.org/10.1007/978-981-19-1332-0

Reference books

1. Li, M., Hammond, G. P., & Musa, H. (2022). Corporate greenhouse gas accounting: Standards, tools, and best practices. Springer. https://doi.org/10.1007/978-3-030-82375-4
2. Matthews, H. S., Hendrickson, C. T., & Matthews, D. H. (2014). Life cycle assessment: Quantitative approaches for decisions that matter.
3. Michael Z. Hauschild, Ralph K. Rosenbaum, Stig Irving Olsen, 2018. Life Cycle Assessment: Theory and Practice, Springer. (Research article)

Online course material

1. Sustainability Counts 2023 Report by PWC and NUS Business School (Centre for Governance and Sustainability): https://www.pwc.com/sg/en/publications/assets/page/sustainability-counts-2023.pdf
2. The Intergovernmental Panel on Climate Change (IPCC): https://www.ipcc.ch/
3. United Nations Framework Convention on Climate Change: https://unfccc.int/
4. GHG Management Institute: https://ghginstitute.org/courses/

Pre-study content

1. Eccles, R. G., Schapiro, M., & Cornell, D. (2022). Accounting for climate change: Measurement, disclosure and corporate practice. Brookings Institution Press.
2. World Resources Institute (WRI), & World Business Council for Sustainable Development (WBCSD). (2004). The greenhouse gas protocol: A corporate accounting and reporting standard (Revised ed.). WRI & WBCSD.

Title	**Resource Resilience**	Number	ETL7XX2
Department	DESE	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech., M.Sc., M.S., M.Tech., Ph.D.	Type	Compulsory
Prerequisite	None	Course Level	700

Objectives

The instructor will:

1. Present a transdisciplinary understanding of resource systems to promote sustainable resourcing and equitable management.
2. Provide students with practical skills in resource planning, environmental assessment, and policy implications.

Learning Outcomes

The students will be able to:

1. Understand and apply system-based approaches to use resources sustainably.
2. Develop strategic responses to emerging resource challenges.
3. Critically evaluate real-world cases and policies in resource use, having strong research, communication, and problem-solving skills.

Contents

Foundational Concepts: Introduction to responsible resourcing and global patterns of usage [4 lectures]

Title	**Systems Approach to Sustainable Development**	Number	ETL7XX3
Department	DESE	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech., M.Sc., M.S., M.Tech., Ph.D.	Type	Compulsory
Prerequisite	Basic understanding of environmental science, engineering systems, or economics is recommended	Course Level	700

Objectives

The instructor will:

1. Introduce systems thinking and modelling approaches for understanding complex sustainability challenges across environmental, technological, and societal domains.
2. Equip students with interdisciplinary tools and metrics to analyze interactions, feedbacks, and trade-offs in real-world integrated systems.

3. Facilitate application of systems-based strategies aligned with institutional and national frameworks (NEP, NAPCC), industry transitions (e.g., circular economy, ESG), and global sustainability goals (UN SDGs, IPCC).

Learning Outcomes

The students will be able to:

1. Apply systems modelling techniques (e.g., stocks-flows, causal loops, feedback analysis) to simulate and interpret complex sustainability systems.
2. Identify the opportunities for creating major impacts with minor interventions.
3. Design and evaluate systems-based solutions for sustainable transitions in sectors like energy, water, waste, or mobility using practical assessment tools and performance indicators.
4. Integrate multi-level policy and industrial frameworks to propose context-specific sustainability strategies grounded in both scientific evidence and stakeholder needs.

Contents

Introduction to Systems Thinking and Sustainability Frameworks: Principles of systems thinking: complexity, feedback loop, non-linearity; Causal loop diagrams, system boundaries; Nexus thinking (energy-water-food-health); Interlinkages with SDGs and national & global goals [7 lectures]

Modelling Tools and Applications: Introduction to system dynamics: stocks, flows, feedback loops; Simulation tools (STELLA, Vensim, and other open-source tools); Scenario planning and model-based policy analysis; Applications: energy, emissions, water, mobility systems [9 lectures]

Climate Systems and Resilience Modelling: Climate risks, adaptation–mitigation linkages; Planetary boundaries, tipping points, system vulnerabilities; Energy-climate-health interactions and policy frameworks (e.g., TCFD) [8 lectures]

Circular Economy and Industrial Sustainability: Lifecycle thinking (using LCA), industrial symbiosis, waste valorisation; Material flow analysis, carbon intensity metrics; Case studies: Energy manufacturing, construction and other sectors [9 lectures]

Multi-Level Governance and Sustainability Policy Instruments: Role of institutions, markets, and communities in sustainability transitions; National and global policies (e.g., India's SDG localization, NAPCC); Systemic instruments (taxes, cap-and-trade, ESG reporting); Participatory systems design [6 lectures]

Textbooks

1. Meadows, D. H. (2008). *Thinking in Systems: A Primer*. Chelsea Green Publishing.
2. Al-Douri, Y. (Ed.). (2022). Renewable Energy: Analysis, Resources, Applications, Management, and Policy. AIP Publishing LLC.
3. Frost, M. (2024). *Environment and Ecology in the Long Nineteenth-Century: Volume II: Popular, Cultural, Social, Political, and Ecological Perspectives on Environment, 1789–1858*. Routledge.

Reference books/articles

1. Kara, S., Hauschild, M., Sutherland, J., & McAloone, T. (2022). Closed-loop systems to circular economy: A pathway to environmental sustainability. *CIRP Annals*, *71*(2), 505-528. (Research article)
2. Fritsch, M., Lischke, H., & Meyer, K. M. (2020). Scaling methods in ecological modelling. Methods in Ecology and Evolution, 11(11), 1368-1378.

3. Khaleel, M. M., Alsharif, A., & Imbayah, I. I. K. (2022). Renewable energy technologies: Recent advances and future predictions. African Journal of Advanced Pure and Applied Sciences (AJAPAS), 58-64. (Research article)
4. Al-Emran, M., & Griffy-Brown, C. (2023). The role of technology adoption in sustainable development: Overview, opportunities, challenges, and future research agendas. Technology in Society, 73, 102240, Elesiver.

Online study material

1. NPTEL course: Energy Resources, Economics and Environment, **2022;** Prof. Rangan Banerjee, IIT Bombay; https://onlinecourses.nptel.ac.in/noc22_hs43/preview**.**
2. NPTEL course: Waste to Energy Conversion, **2025;** Prof. P. Mondal, IIT Roorkee; https://onlinecourses.nptel.ac.in/noc25_ch46/preview**.**
3. Circular Economy Toolkit: Ellen MacArthur Foundation https://www.ellenmacarthurfoundation.org/articles/circular-buildings-toolkit

Title	**Environmental Technologies**	Number	ETL7XX1
Department	DESE	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech., M.Sc., M.S., M.Tech., Ph.D.	Type	Compulsory
Prerequisite	None	Course Level	700

Objectives

The instructor will:

1. Introduce a range of traditional & contemporary technologies for sustainable development.
2. Emphasizing their practical applications in addressing real-world societal and environmental challenges and their linkage with SDGs.

Learning Outcomes

The students will be able to:

1. Understand the linkage between technologies and sustainable development.
2. Identify, assess, and apply appropriate technologies to address specific sustainability challenges in areas such as water, energy, waste, air, infrastructure, and carbon management.
3. Evaluate and compare the implications of and develop upstream and downstream technological solutions for industrial situations as well as at local, national, and global scales.

Contents

Introduction: Traditional & contemporary technologies for sustainable development; Front-end and back-end technologies and their contribution to a circular economy [5 lectures]

Front-End Sustainable Technologies: Technologies that enable sustainable design and prevention (modelling and designed with case study) [17 lectures]

Design of sustainable products, cleaner production, demand-side resource efficiency and infrastructure including, Urban Design (AUTOCAD; ReVIT; EPANET; openLCA; GREET): Eco-design, biomimicry, modular design, low-impact materials & innovations; Water-saving devices, energy-efficient appliances; Lean manufacturing, zero-waste production; Passive cooling, smart green cities, sustainable transportation

Back-End Sustainable Technologies: Technologies that manage, treat, and remediate environmental impacts [17 lectures]

Design of pollution control technologies (EPANET; LandGEM; SewerGEM; HYDRUS/ DWSIM): Air scrubbers, effluent treatment plants (ETPs), solid waste treatments, constructed

wetlands, and faecal sludge treatment plants (FSTPs); Bioremediation and phytoremediation; Design of Carbon Capture Utilisation & Storage (CCUS)

Textbooks

1. Nhamo, G., Chikodzi, D., & Dube, K. (2021). Sustainable Development Goals for Society Vol. 1. Springer International Publishing.
2. Book chapter (E-Reserves): C.U. Becker, Sustainability Ethics and Sustainability Research, Ch. 2. Meaning of Sustainability, Springer 2012, pages 9-15.

Reference books

1. Sachs, J., Schmidt-Traub, G., Kroll, C., Lafortune, G., Fuller, G., and Woelm, F., 2020. The Sustainable Development Goals and COVID-19. Sustainable Development Report 2020. Cambridge: Cambridge University Press.
2. Prasad, M. N. V. (Ed.). Sustainable Development Goals: Perspectives, Challenges and Solutions in India. Springer Nature, Singapore, 2021. ISBN: 978-981-15-7196-3
3. Somnath Hazra and Anindya Bhukta., 2020, Sustainable Development Goals: An Indian Perspective, 1st edition, Springer International Publishing, ISBN: 978-3-030-42490-9.

Online Learning material

1. United Nations SDG Book Club: https://www.un.org/sustainabledevelopment/sdgbookclub/
2. United Nations SDG Goals: https://sdgs.un.org/goals
3. CSR Online Certification Course (Education Nest): https://educationnest.com/courses/csr-online-certification-training-course
4. NITI Aayog – SDG India Index: https://www.nitiforstates.gov.in/
5. SDG Academy: https://sdgacademy.org/
6. Sachs, Jeffrey D. *The Age of Sustainable Development*. Columbia University Press, New York, 2015. https://cup.columbia.edu/book/the-age-of-sustainable-development/9780231173155/

Pre-study contents

Introduction, definition, and Principles of Sustainable Development; Aspects of Sustainability, and the Transition from Millennium Development Goals (MDGs) to Sustainable Development Goals (SDGs); Brundtland Commission report; Millennium ecosystems assessment; IPCC reports, evolution from green engineering to sustainable engineering; The need for SDGs, interlinkages between SDGs and their global adoption; and Role of science and technology; Conventional and sustainable development; Circular economy; Design for Sustainability (D4S); Thinking alternatives and innovation; Multi-stakeholder mapping; Dynamic and interrelated nature of social-environmental systems; and Sustainability quantifications [7 lectures]

Title	**Environmental and Sustainability Engineering Laboratory**	Number	ETL7XX1
Department	DESE	L-T-P [C]	0-0-4 [2]
Offered for	M.Tech.	Type	Compulsory
Prerequisite	None	Course Level	700

Objectives

The instructor will:

1. To develop practical understanding of sustainability assessment tools and techniques for evaluating environmental interventions.
2. To provide hands-on experience with environmental monitoring and characterization techniques.
3. To enable students to apply integrated sustainability assessment approaches to real-world problems.

Learning Outcomes

The students will be able to:

1. Apply sustainability assessment methods to environmental systems.
2. Perform laboratory and field-based environmental analyses.
3. Conduct integrated sustainability assessments aligned with real-world needs.

Contents

Sustainability Assessment Tools and Techniques

1. Carbon and water footprint analysis: A case study.
2. Water–energy–carbon nexus analysis using scenario-based modelling.
3. Urban or regional sustainability indicators assessment mapped to SDGs.
4. Multi-criteria decision analysis (MCDA) for sustainability alternatives.
5. Cost-benefit and cost-effectiveness analysis of sustainability interventions.
6. Capstone lab project (Lab 6-10): Integrated sustainability assessment of a real-world system.

Environmental Technologies

1. Water quality analysis (pH, turbidity, conductivity, TDS, TSS, Odor).
2. Wastewater biochemical characterization (BOD/COD/TKN/TP/ Total coliforms).
3. Estimating energy consumption for energy audit.
4. Carbon footprint estimation.
5. Solid waste characterization (Biodegradability, density, viscosity, moisture analysis).
6. Waste to energy conversion- understanding anaerobic digestion set up.
7. Ambient and indoor air quality monitoring and particulate characterization.
8. Microclimate and heat stress analysis using field measurements.
9. Field Immersion: Community engagement for identification and formulation of real-world sustainability problem statements.

Textbooks

1. Nhamo, G., Chikodzi, D., & Dube, K. (2021). Sustainable Development Goals for Society Vol. 1. Springer International Publishing.
2. Mihelcic, J. R., & Zimmerman, J. B. (2014). Environmental Engineering: Fundamentals, Sustainability, Design (2nd ed.). John Wiley & Sons.

Reference books

1. Allen, D. T., & Shonnard, D. R. (2011). Sustainable Engineering: Concepts, Design and Case Studies. Prentice Hall.
2. Hauschild, M. Z., Rosenbaum, R. K., & Olsen, S. I. (2018). Life Cycle Assessment: Theory and Practice. Springer International Publishing.

Program Elective Courses

Title	**Water Sustainability**	Number	ETL7XX X
Department	DESE	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech., M.Sc., M.S., M.Tech., Ph.D.	Type	Elective

Prerequisite	None	Course Level	700

Objectives

The instructor will:

1. Reflect on the fundamentals of science and technology associated with sustainability of water.
2. Address the area-specific problems in water for achieving UN's sustainable development goal (SDG) 6 of *Clean Water and Sanitation and* SDG 14 of *Life below Water*.
3. Reflect on the sustainability of water by reducing, reusing, and recycling of water efficiently in different sectors.

Learning outcomes

The students are expected to have the ability to:

7. Understand the science and technologies related to water in different contexts.
8. Critically evaluate the necessity of water sustainability and factors affecting it.
9. Develop water alertness, which in turn will help to spread awareness about safe water use, reuse, and conservation.

Contents

Fundamentals of Water Systems and Global Water Challenges: Hydrological cycle and water resources; Water availability and demand trends: global and Indian perspectives; The Indian water scenario: challenges and regional disparities; Water and the SDGs; Water as a human right: legal, ethical, and humanitarian perspectives; and Key government interventions [4 lectures]

Water Quality and Health Implications: Water quality parameters, drinking water standard: WHO and BIS guidelines; Water quality benchmarks for irrigation, drinking, and industrial use; and Water and health: essential minerals in water, emerging pollutants, and health impact of water treatment technologies [8 lectures]

Water Treatment, Reuse, and Technological Innovations: One water: integrated water management and circular water economy; Conventional treatment technologies, nature-based solutions, advanced & emerging technologies: membrane filtration, capacitive deionization, ion exchange membranes, and desalination technologies; Distillation and thermal processes; Hybrid water treatment systems; Wastewater reclamation, treatment, and safe reuse practices; International case study: Israel's water reuse and efficiency model; and Field component: visit to a water or wastewater treatment plant [15 lectures]

Water Management and Efficiency: Major water use sectors; Water conservation strategies; Smart water management; Water-Energy-Food nexus; Water footprint; Water credit; and Case studies: water efficiency in agriculture, industries, and municipal sectors, zero liquid discharge, and injection wells [12 lectures]

Textbooks

10. J. A. A. Jones, *Water Sustainability: A Global Perspective,* First Edition, Routledge, Taylor & Francis Group, 2015, ISBN: 9781138167940
11. P.K. Tewari, *Advanced Water Technologies: Concepts and Fundamentals*, CRC Press, Taylor and Francis Group, 2020, ISBN: 9781315101514
12. Nicholas Gray, *Water Science and Technology: An Introduction*, Fourth Edition, CRC Press, 2017, ISBN: 9781498753456

Reference books

1. Simon J. Dadson, Dustin E. Garrick, Edmund C. Penning-Rowsell, Jim W. Hall, Rob Hope, Jocelyne Hughes, *Water Science, Policy, and Management: A Global Challenge*, 2019, John Wiley & Sons Ltd., ISBN:9781119520603
2. Frank R. Spellman, *The Science of Water: Concepts and Applications*, Fourth Edition, CRC Press, 2020, ISBN: 9780367555887

Online study material

1. Prof. Jenia Mukherjee, Water, Society and Sustainability, **2018**, NPTEL course material, Department of Humanities and Social Sciences, Indian Institute of Technology Kharagpur, https://nptel.ac.in/courses/109105136

2. Prof. Bhanu Prakash Vellanki, Water and Wastewater Treatment, **2018**, NPTEL course material, Department of Civil Engineering, IIT Roorkee, https://nptel.ac.in/courses/105107207

Title	**Principles of Membrane Fabrication**	Number	ETL7XXX
Department	DESE	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech., M.Sc., M.S., M.Tech., Ph.D.	Type	Compulsory
Prerequisite	None	Course Level	700

Objectives

The instructor will:

1. Reflect on the fundamentals of science and technology associated with membrane fabrication.
2. Provide an insight to the conventional and emerging membrane technologies.
3. Address key aspects to achieve energy-efficient separations through membranes.

Learning outcomes

The students are expected to have the ability to:

1. Apply the factors affecting the selection, design, and operation of membrane separation processes, for energy-efficient and sustainable separations.
2. Design appropriate membrane-based separation technique(s) as per the need.
3. Describe process parameters and key factors governing membrane structures.

Contents

Introduction to Membrane Science and Technology: Membranes; Historical development of membranes; Membrane applications related to gas and liquid separation; Membrane materials; Types of membranes; Membranes geometries: flat sheets, hollow fiber, and tubular membranes; Membrane modules: plate-and-frame, spiral-wound, tubular, and hollow fiber modules; and Membranes and circular economy [10 lectures]

Membrane Fabrication Processes: Casting of flat sheet membranes and phase inversion process; Spinning of hollow fiber membranes; Roll-to-roll coating for composite membranes; Thin-film composite membranes and interfacial polymerization; Sintering, track-etching, and leaching techniques [14 lectures]

Parameters to Control Membrane Characteristics: Processing parameters; Raw materials, additives, polymer solution characteristics, and ambient conditions (temperature and humidity) [5 lectures]

Membrane Post-Treatments: Atomic Layer Deposition (ALD) and Chemical Vapor Deposition (CVD) on membrane surfaces; Chemical treatment; and Radiation treatment [4 lectures]

Emerging Membrane Technologies: Isoporous membranes; Charge mosaic membranes, Biological membranes; Ion exchange membranes, Electroconductive membranes, and Membranes using functional material [6 lectures]

Textbooks

1. Richard W. Baker, *Membrane Technology and Applications*, 3rd Edition, John Wiley & Sons, Ltd., 2012, ISBN: 978-0470743720
2. Heinrich Strathmann, *Introduction to Membrane Science and Technology*, Wiley-VCH Verlag GmbH, 2006, ISBN: 9783527324514

Reference books

1. P. K. Tewari, *Nanocomposite Membrane Technology: Fundamentals and Applications*, CRC Press, Taylor & Francis Group, 2016, ISBN: 9781138749122
2. Marcel Mulder, *Basic Principles of Membrane Technology*, 2nd Edition, Springer Dordrecht, 2012, ISBN: 9789400917668
3. Takeshi Matsuura, *Synthetic Membranes and Membrane Separation Processes,* Japan: CRC-Press, 1994, ISBN: 9780849342028

Online study material

1. Kaustubha Mohanty, Membrane Technology, **2019**; NPTEL course material, Department of Chemical Engineering, IIT Guwahati, https://nptel.ac.in/courses/103103163/

Title	**Industrial Waste Management and Circular Economy**	Number	ETL7XXX
Department	DESE	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech., M.Sc., M.S., M.Tech., Ph.D.	Type	Compulsory
Prerequisite	None	Course Level	700

Objectives

The instructor will:

1. Explain the importance of waste management in the Industrial sector.
2. Explore the key challenges and barriers to the circular economy and the concept of 3R's.
3. Teach about different circular designs and innovations.

Learning outcomes

The students are expected to have the ability to:

1. Understand the different types of waste generation in industrial areas.
2. Understand the key challenges and barriers to the circular economy and the concept of 3R's
3. Design of sustainable circular design and innovations.

Contents

Introduction to the type of waste in Industries: Source of industrial waste; and Types of industrial waste: liquid and solid waste [3 lectures]

Characteristics of industrial waste: Inorganic salts; Acids/ or alkalines; Organic matter; Toxic chemicals; Radioactive materials; Foam forming matter; Hazardous material; Hazard identification; Risk analysis; and Methods of industrial waste disposal: Gasification, Pyrolysis, Anaerobic digestion for sludges, and Landfills [7 lectures]

Industrial waste management: Industrial wastewater discharges into streams; Lakes and oceans and problems; Recirculation of industrial wastes; Use of municipal wastewater in industries [7 lectures]

Circular economy: Material recovery; Waste reduction; Butterfly diagram; Process changes; Equipment modifications; Segregation of wastes; Equalization of wastes; By-product recovery;

Proportioning of wastes; Monitoring waste streams; and Product cycle [10 lectures]
Circular design, innovation, and assessment: Concept of loops; Zero waste: Waste management in context of circular economy; Circular design; Research and innovation; Life cycle analysis; and Circular business models [8 lectures]
Case Studies: EPR: polluters pay principle, and Industrial symbiosis/ Eco-parks [4 lectures]
Textbook

1. M. N. Rao and A. K. Datta, *Wastewater Treatment: Rational Methods of Design and Industrial Practices*, 2nd ed. Oxford, U.K.: Oxford & IBH Publishing, 2017.
2. H. Tchobanoglous, F. L. Burton, and G. Tchobanoglous, *Wastewater Engineering: Treatment and Resource Recovery*, 5th ed. New York, NY, USA: McGraw-Hill Education, 2013.

Reference Books

1. L. F. Diaz, M. de Bertoldi, W. Bidlingmaier, and E. Stentiford, *Compost Science and Technology*, 1st ed. Amsterdam, Netherlands: Elsevier, 2007.
2. K. E. Martin, *Industrial Waste Management and Cleaner Production*, 1st ed. New Delhi, India: CBS Publishers & Distributors, 2016.

Online Learning material

1. U.S. Environmental Protection Agency (USEPA), *2023 Industrial Waste Management Guide to Best Practices*. National Service Center for Environmental Publications, 2023.
2. S. Peters and K. Samarasinghe, "Waste Management Practices: Innovation, Waste to Energy and e-EPR," in *An Introduction to Circular Economy*, L. Liu and S. Ramakrishna, Eds. Singapore: Springer, 2021. [Online]. Available: https://doi.org/10.1007/978-981-15-8510-4_15

Title	**Introduction to ESG regulation, reporting, and accounting**	Number	ETL7XXX
Department	DESE	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech., M.Sc., M.S., M.Tech., Ph.D.	Type	Elective
Prerequisite	None	Course Level	700

Objectives
The instructor will:
1. Explain the objectives and importance of Environmental, socioeconomic, and Governance
2. Teach about the various ESG frameworks and standards and how to apply them to your context and goals.

Learning outcomes
The students are expected to have the ability to:
1. Understand the importance and benefits of ESG integration and reporting for your organization and stakeholders.
2. Learn about the various ESG frameworks and standards and how to apply them to your context and goals.

Contents
Introduction to ESG: Definition and concept of ESG, Evolution of ESG in business and finance, Importance and objectives of ESG, Impact on financial performance and risk management [4 lectures]
Environmental (E) Factors: Understanding climate risk, Greenhouse gas emissions and reduction strategies, Water usage and Waste management, Assessing and preserving biodiversity [6 lectures]

Social (S) Factors: Diversity and inclusion, Labor practices and human rights, Responsible sourcing, Product safety and quality, Customer privacy and data protection [4 lectures]
Governance (G) Factors: Corporate Governance, Ethics and Anti-Corruption, Risk Management and Compliance [4 lectures]
Introduction to ESG Regulations; Overview of global, regional, and local ESG regulations; Regulatory bodies and their roles; Key ESG Reporting Frameworks; Core elements and indicators [6 lectures]
ESG Accounting Principles; ESG Accounting vs. Traditional Accounting; Integrating ESG metrics into financial reporting; Materiality and ESG Accounting; ESG Assurance and Verification; Role of Assurance in ESG Reporting [8 lectures]
ESG Reporting and Standards: Global Reporting Initiative, Sustainability Accounting Standards Board, Task Force on Climate-related Financial Disclosures [7 lectures]
Textbooks
1. Blokdyk, Gerardus (2019), GHG Emissions Accounting A Complete Guide 2019 Edition, Practical Tools for Self Assessment, ISBN-10 : 0655542779, ISBN-13 : 978-0655542773
2. Muthu, S. S. (Ed.). (2022). Greenhouse gas accounting: Global frameworks and emerging issues. Springer. https://doi.org/10.1007/978-981-19-1332-0

Reference Books
1. McDonough, W., & Braungart, M. (2013). The Upcycle: Beyond Sustainability – Designing for Abundance. North Point Press.
2. Gates, B. (2021). How to Avoid a Climate Disaster: The Solutions We Have and The Breakthroughs We Need. Knopf Publisher.

Online Learning material
1. The Intergovernmental Panel on Climate Change (IPCC): https://www.ipcc.ch/
2. United Nations Framework Convention on Climate Change: https://unfccc.int/
3. GHG Management Institute: https://ghginstitute.org/courses/

Title	**Technologies for Rural Sanitation**	Number	ETL7XXX
Department	DESE	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech., M.Sc., M.S., M.Tech., Ph.D.	Type	Elective
Prerequisite	None	Course Level	700

Objectives
The instructor will:
1. Explain the importance of sanitation in rural areas and its impact on public health and well-being.
2. Teach about different sanitation technologies and approaches suitable for rural areas.
3. Explain the critical design aspects of sanitation technologies for rural areas.

Learning outcomes
The students are expected to have the ability to:
1. Listen, comprehend, and value the inherent diversity of communities concerning sanitation approaches.
2. Learn the different designs of sustainable sanitation technologies for rural.
3. Students should showcase improved sanitation technology designs for rural areas that can be climate-resistant and resilient.

Contents
Introduction to Sanitation in Rural Areas: Understand the importance of sanitation in rural

areas. Explore the key challenges and barriers to sanitation in rural communities. Liquid waste management challenges in rural communities. Liquid waste management strategies and practices. Recycling and resource recovery options [6 lectures]

Sanitation Technologies for Rural Areas: Different sanitation technologies suitable for rural communities. Design of sanitation technologies like

I. Pit latrines- Load calculation, flow systems, characteristics of sludge produced in the pit systems, and safe disposal

II. Septic tanks- Load calculation, flow system, characteristics of sludge produced in the pit systems, and safe disposal

II. Bio-toilets- Load calculation, flow system, characteristics of sludge produced in the pit systems, and safe disposal

IV. Dry toilets (Ecosan); Load calculation, flow system, characteristics of sludge produced in the pit systems, and safe disposal [18 lectures]

Design of the sanitation for a village: Load calculation for a small town, faecal sludge production method, faecal sludge collection method, faecal sludge characterisation; factors affecting the faecal sludge load calculation; Physical-biological-chemical-nutrient properties of faecal sludge and treatment methods [10 lectures]

Case Studies and Best Practices: Examining successful rural sanitation projects in India and worldwide. Lessons learned and practical insights from case studies [5 lectures]

Textbook

1. G. S. Birdie and J. S. Birdie, *Water Supply and Sanitary Engineering*, 8th ed. New Delhi, India: Dhanpat Rai Publishing Company, 2015.
2. L. Strande, M. Ronteltap, and D. Brdjanovic, *Faecal Sludge Management: Systems Approach for Implementation and Operation*, 1st ed. London, U.K.: IWA Publishing, 2014.

Reference Books

1. Lucena, J., Schneider, J., and Leydens, J.A., Engineering and Sustainable Community Development. Morgan and Claypool Publishers, 2010. 1st Edition.
2. E. Tilley, L. Ulrich, C. Lüthi, P. Reymond, and C. Zurbrügg, *Compendium of Sanitation Systems and Technologies*, 2nd ed. Dübendorf, Switzerland: Eawag (Swiss Federal Institute of Aquatic Science and Technology), 2014.
3. R. Franceys, J. Pickford, and R. Reed, *A Guide to the Development of On-site Sanitation*. Geneva, Switzerland: World Health Organization, 1992.

Online Learning material

1. Solar Energy International, "Appropriate Technology for the Developing World – Online," *Solar Energy International (SEI)*. [Online]. Available: https://www.solarenergy.org/courses/appropriate-technology-for-the-developing-world-online-2/. [Accessed: May 27, 2025].
2. World Health Organization, *Guidelines on Sanitation and Health*, 2018. [Online]. Available: https://www.who.int/water_sanitation_health/publications/guidelines-onsanitation-and-health/en/. [Accessed: May 27, 2025].

Title	**Biodiversity and Natural Resources**	Number	ETL7XX X
Department	DESE	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech., M.Sc., M.S., M.Tech., Ph.D.	Type	Elective

Prerequisite	None	Course Level	700

Objectives

The instructor will:

1. To equip participants with a deep understanding of biodiversity and its intrinsic value, including its role in sustaining ecosystems and human well-being.
2. To analyze in detail the impact of resource depletion and biodiversity loss on various sectors and society, emphasizing the role of human activities as key drivers.

Learning Outcomes

The students are expected to have the ability to:

1. Critically examine the major threats to biodiversity and natural resources (BNR), with a strong focus on human activities.
2. Explore the various types and complexities of natural resources and their importance for human well-being and economic development.
3. Develop competency in sustainable practices and policy frameworks for conservation and resource management.

Contents

Introduction to BNR – Concept, scope and constraints of biodiversity and natural resource; Ecosystem services, ecological footprint; Biodiversity at global, national and local level; Natural resources of different geographical regions and their Classification [6 lectures]

Challenges and Threats to BNR –Drivers of habitat loss, overexploitation of resources, pollution, climate change; Biodiversity Indices; Factors influencing natural resource availability, distribution and uses; Impact of resource depletion on different sectors [8 lectures]

Technologies and Sustainable Practices: Technologies for natural resource management (NRM) – Management of common national and international resources and their management commissions; *Case studies from few agro-ecological zones at global and national level*; Integrated NRM; Community-based NRM; Building sustainable business; Case Studies of industry leaders in sustainable practices [12 lectures]

Quantification and Assessment of BNR: Biodiversity Valuation and Ecosystem Services Accounting; Approaches in resource management; Natural resource assessment; Natural resource accounting and auditing; Real-world examples of quantification methods in leading companies [13 lectures]

Textbooks

1. Shivakoti, G., Pradhan, U., & Helmi, H. (Eds.). (2016). Redefining diversity and dynamics of natural resources management in Asia, volume 1: sustainable natural resources management in dynamic Asia. Elsevier. ISBN: 978-0-12-805454-3.
2. Groom, M. J., Meffe, G. K., Carroll, C. R., & Andelman, S. J. (2006). Principles of conservation biology (Vol. 3). Sunderland: Sinauer associates.

Reference Books

1. Lynch, D. R. (2009). Sustainable natural resource management: For scientists and engineers. Cambridge University Press.
2. Primack, R. B. (2006). Essentials of conservation biology (Vol. 23). Sunderland: Sinauer Associates.
3. Dash, M. C. (2013). Concepts of Environmental Management for Sustainable Development. IK International Pvt Ltd.

Online Learning material

1. NPTEL, "Agricultural Development and Rural Transformation," [Online]. Available: https://onlinecourses.nptel.ac.in/noc22_ag10/preview.
2. SWAYAM, "Gender Justice and Workplace Security," [Online]. Available: https://onlinecourses.swayam2.ac.in/cec21_ge31/preview.

Title	**Climate change Risk, Adaptation and Mitigation**	Number	ETL7XXX
Department	DESE	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech., M.Sc., M.S., M.Tech., Ph.D.	Type	Elective
Prerequisite	None	Course Level	700

Objectives

The instructor will:

1. Discuss the current issues of climate and climate change at local, national, and global levels.
2. Examine the scientific aspects of climate change and its implications.
3. Analyze strategies for climate change adaptation and mitigation at various sectors.

Learning Outcomes

The students are expected to have the ability to:

1. Identify evidence-based drivers and mechanisms for climate change and environmental disasters.
2. Explore the concepts and attributes of adaptation and mitigation to minimize the impacts of climate change.
3. Demonstrate the various research methods/assessment techniques to evaluate the adverse impacts of climate change.

Contents

Introduction to Climate and Climate Change– Ecology and Environment; Weather and Climate; Climate system and Categories; Climate and Climate change: *Risk, Hazard and Vulnerability* [4 lectures]

Impacts of Changing Climate – *Global Trends on various impacts*: Heatwaves; Rising Sea Levels; Flood and Drought; Wildfires; Precipitation and Storms; Atmospheric Pollution; Allergens and Pollen; Vector-borne Diseases; Food and Water-borne Diseases; Food Security; **Energy Crisis**; *Case studies*: Region-specific impacts; Nation-specific impacts [10 lectures]

Adaptation Strategies: Concepts and Attributes of Climate Change Adaptation; Risk-Response Framework at Global, National, Local and Sectoral level; Preparedness and Response Strategies; Key Initiatives; Case Studies-Incorporating TCFD and Best Practices. [8 lectures]

Mitigation Strategies: Short, Medium, and Long-term Mitigation Goals; Mechanisms of Sustainable Development; Current Controversies and Debates; Greenhouse Gas Emission Trends and Global Debates on Cost-Sharing; Policies, Measures, and Instruments; Case Studies-Incorporating ERM [8 lectures]

Quantitative Assessment Techniques: Sources and Inventories of Weather/Climate Variables; Projections and Forecasting of Climate Change; Sources of Climate Outcomes in Retrospective Time Series; Statistical Methods; Analytical and Data Challenges [9 lectures]

Textbooks

1. Qu, J. J., & Motha, R. P. (2022). Climate Change and a Sustainable Earth. Cambridge Scholars Publishing, ISBN 978-1-527-58044-2.

2. Carlsten, C., Jaakkola, J. J., Bayram, H., Sigsgaard, T., Elci, M. A., & Costa, D. L. (2021). *Climate change and global public health*. K. E. Pinkerton, & W. N. Rom (Eds.). Humana Press, ISBN 978-3-030-54746-2.

Reference Books

1. Uitto, J. I., Puri, J., & Van den Berg, R. D. (2017). Evaluating climate change action for sustainable development. Springer Nature.
2. Kovats, S., Ebi, K. L., & Menne, B. (2003). *Methods of assessing human health vulnerability and public health adaptation to climate change* (No. EUR/03/5036776). World Health Organization. Regional Office for Europe.

Online Learning material

1. "Introductory e-Course on Climate Change,".-https://www.uncclearn.org/courses/introductory-e-course-on-climate-change.
2. NextGenU.org, "Climate Change and Health Course,"-https://courses.nextgenu.org/course/view.php?id=307
3. Intergovernmental Panel on Climate Change (IPCC),-https://www.ipcc.ch/.

Title	**Industrial Ecology and Life Cycle Assessment**	Number	ETL7XXX
Department	DESE	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech., M.Sc., M.S., M.Tech., Ph.D.	Type	Compulsory
Prerequisite	None	Course Level	700

Objectives

The instructor will:

1. To realize the importance of sustainability and its quantification.
2. To apply the principles of industrial ecology to improve sustainability.
3. To apply the concept of life cycle assessment (LCA) for evaluating environmental impacts.

Learning outcomes

The students are expected to have the ability to:

1. Appreciate the various concepts of industrial ecology.
2. Utilize concepts of industrial ecology to develop preliminary solutions for sustainability problems.
3. Formulate and solve simple LCA problems.
4. Analyze and interpret LCA results to provide recommendations.

Contents

Overview

Sustainability – definition, three dimensions, sustainable development goals [2 lectures]

Need for sustainability quantification, footprint analysis, economic indicators, social indicators, safety and sustainability [3 lectures]

Industrial Ecology

Definition and principles of industrial ecology, circular economy, carrying capacity, IPAT equation, scales of influence [4 lectures]

Analogy between industrial systems and biological systems, case studies (3 *lectures*)

Concepts and tools - cleaner production, industrial symbiosis, biomimicry, material flow analysis, input output analysis [7 lectures]

Life Cycle Assessment (20 lectures)

Concept of life cycle thinking, goal and scope, life cycle inventory, impact assessment methods, characterization factors [8 lectures]

Introduction to software packages for LCA, inventory databases, midpoint and endpoint indicators, cultural perspectives in LCA, hands-on activities [8 lectures]

Attributional and consequential LCA, case studies, limitations of LCA [5 lectures]

Textbooks

1. T. E. Graedel and Braden Allenby, 2009. Industrial Ecology and Sustainable Engineering, Pearson.
2. Adisa Azapagic and Slobodan Perdan, 2011. Sustainable Development in Practice: Case Studies for Engineers and Scientists, Wiley.

Reference books

13. David Brennan, 2012. Sustainable Process Engineering: Concepts, Strategies, Evaluation and Implementation, Pan Stanford.
14. Michael Z. Hauschild, Ralph K. Rosenbaum, Stig Irving Olsen, 2018. Life Cycle Assessment: Theory and Practice, Springer.

Online study material

1. B. K. Dubey, Sustainable Engineering Concepts and Life Cycle Analysis, NPTEL course material, https://nptel.ac.in/courses/105105157 OpenLCA Open Source Software, www.openlca.org
2. The International Society for Industrial Ecology, www.is4ie.org
3. The Forum for Sustainability through Life Cycle Innovation, https://fslci.org/
4. IchemE The Institution of Chemical Engineers, The Sustainability Metrics, https://www.icheme.org/media/16021/sustainability-metrics.pdf